

Highlights

- In text-based person re-identification, we introduce **Bidirectional Local-Matching (BiLMa) framework**. Direction means:
 - image2text**: align textual and visual parts via solving MLM.
 - text2image**: align textual and visual parts via solving MLM.
- We propose a novel MIM method, **Semantic MIM (SemMIM)**, to narrow the semantic gap between person images and captions. We formulate SemMIM as a prediction of semantic labels for randomly masked image patches.
- We show our BiLMa w/ SemMIM can improve performances on three benchmark datasets.

Problem Statement

- Text-based person re-identification (TBPreID) aims to retrieve a target person from an image pool given a textual query.



Future Work

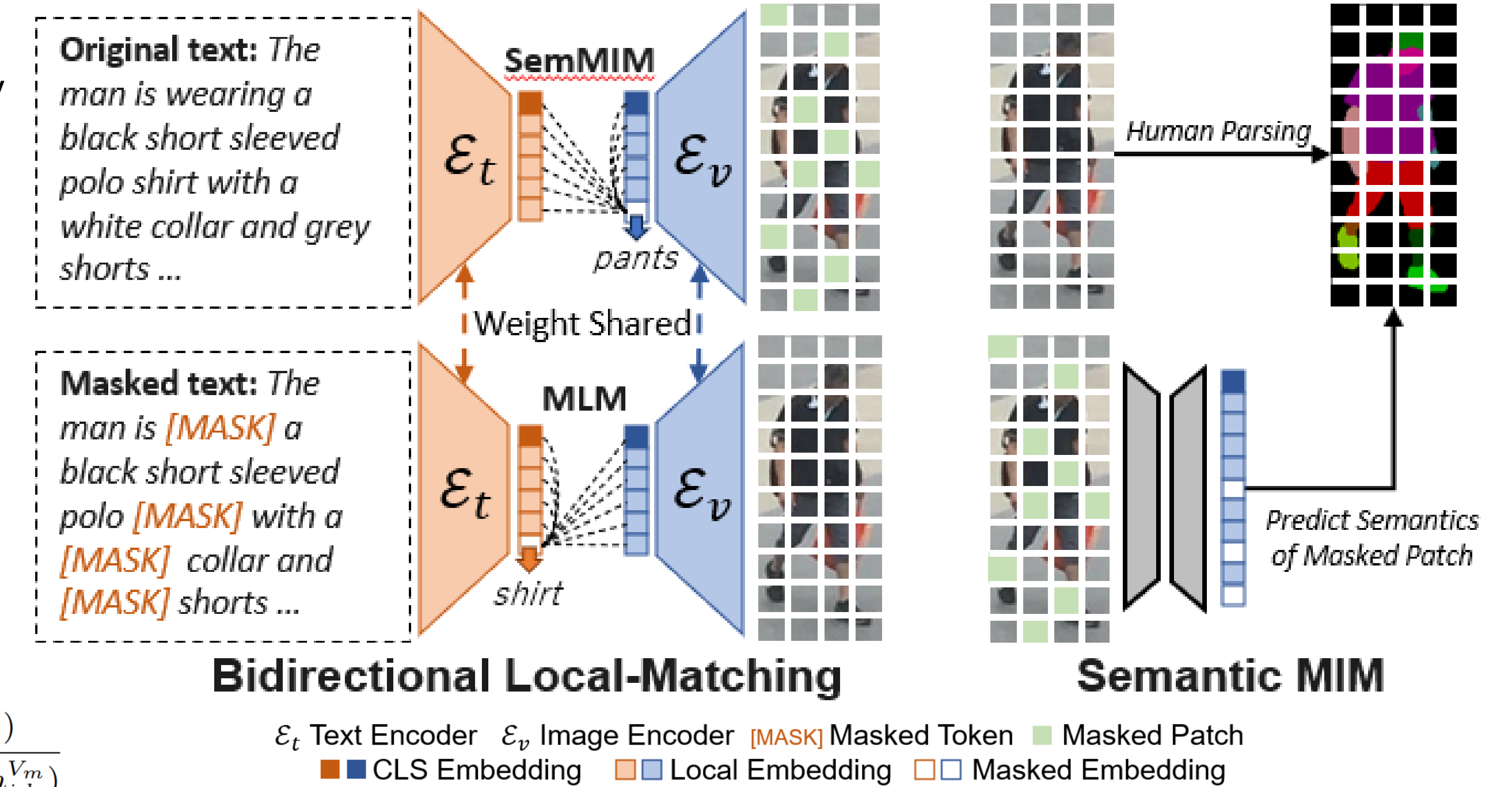
- We will find more helpful Masked Image/Language Modeling strategies for text-based person re-identification.
- We will investigate the influence of human parser's errors and explore ways to alleviate them.

Reference:

[1] Peike Li, Yunqiu Xu, Yunchao Wei, and Yi Yang. Self-correction for human parsing. IEEE Transactions on Pattern Analysis and Machine Intelligence, 2020.

Method

- Bidirectional Local-Matching (BiLMa)** framework jointly optimizes Masked Language Modeling (MLM) and Masked Image Modeling (MIM).
- Input images and texts are randomly masked, and a model predicts masked textual tokens and visual patches by all the unmasked ones.
- Overall loss: $\mathcal{L} = \underbrace{\mathcal{L}_{id} + \mathcal{L}_{sdm}}_{\text{Global Matching}} + \underbrace{\alpha\mathcal{L}_{mlm} + \beta\mathcal{L}_{mim}}_{\text{Bidirectional Local Matching}}$
- Semantic MIM (SemMIM)** is formulated MIM as a prediction of semantic labels for masked image patches.
- Semantic labels of masked image patches are automatically given by a SOTA human parser [1].
- SemMIM loss: $\mathcal{L}_{mim} = -\frac{1}{|\mathcal{M}_v||\mathcal{C}|} \sum_{i \in \mathcal{M}_v} \sum_{j \in |\mathcal{C}|} y_j^i \log \frac{\exp(m_{i,j}^{V_m})}{\sum_{k=1}^{|\mathcal{C}|} \exp(m_{i,k}^{V_m})}$



Experimental Results

- Comparison with Baselines:**
- BiLMa achieves the best Rank@1 and mAP on all benchmark datasets.
- Particularly, compared to baselines,
 - CUHK-PEDES**: 0.56% higher on R@1
 - ICFG-PEDES**: 0.37% higher on mAP
 - RSTPreid**: 1.00% and 1.34% higher on R@1 and mAP.

Method	CUHK-PEDES				ICFG-PEDES				RSTPreid			
	R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP
ISANet [24]	63.92	82.15	87.69	-	57.73	75.42	81.72	-	-	-	-	-
LBUL [21]	64.04	82.66	87.22	-	-	-	-	-	45.55	68.20	77.85	-
AXM-Net [7]	64.44	80.52	86.77	58.73	-	-	-	-	-	-	-	-
LGUR [17]	65.25	83.12	89.00	-	59.20	75.32	81.56	-	-	-	-	-
IVT [19]	65.59	83.11	89.21	-	56.04	73.60	80.22	-	46.70	70.00	78.80	-
CFine [23]	69.57	85.93	91.15	-	60.83	76.55	82.42	-	50.55	72.50	81.60	-
IRRA [10]	73.38	89.93	93.71	66.13	63.46	80.25	85.82	38.06	60.20	81.30	88.20	47.17
BiLMa w/ SemMIM (Ours)	74.03	89.59	93.62	66.57	63.83	80.15	85.74	38.26	61.20	81.50	88.80	48.51

Ablation Study on Each Component:

- Both MLM and SemMIM (i.e., BiLMa) tend to achieve the best performance, indicating the good compatibility of SemMIM with MLM.
- Interestingly, in several cases, our model with only SemMIM outperforms the model with only MLM, which implies strong ability of SemMIM

Components		CUHK-PEDES				ICFG-PEDES				RSTPreid			
MLM	SemMIM	R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP	R@1	R@5	R@10	mAP
✓		73.01	88.92	93.58	65.62	63.09	80.00	85.62	37.99	59.50	80.55	88.35	47.06
	✓	73.16	89.52	93.63	66.00	63.60	80.29	85.70	38.12	59.05	80.35	87.95	46.29
✓	✓	73.55	89.41	93.54	66.28	63.08	80.11	85.63	37.97	59.40	80.70	87.35	46.05
✓	✓	74.03	89.59	93.62	66.57	63.83	80.15	85.74	38.26	61.20	81.50	88.80	48.51